

Vishva Niral Patel

B.Tech Electronics and Communication Engineering

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Research Interests

Artificial Intelligence, Machine Learning, Computer Vision, Biometric Authentication Systems, Secure AI Systems, Intelligent Embedded Systems, Autonomous and Real-World Intelligent Systems.

Education

2023 – 2027 **Bachelor of Technology (B.Tech)**, *Electronics and Communication Engineering*, Dharmsinh Desai University, India
CGPA: 9.13 / 10 (till Semester 5)

Relevant Coursework

Signals and Systems
Digital Electronics
Control Systems
Microprocessors and Embedded Systems

Projects

Multi-Stage Biometric Authentication and Reality Verification System (DARVS)

- Designed a multi-stage biometric authentication framework integrating face recognition, liveness detection, and system-level integrity verification.
- Developed a Python-based authentication pipeline with a browser-based camera interface for real-time biometric verification.
- Proposed a security architecture separating reality verification from biometric recognition to mitigate replay, spoofing, and deepfake attacks.

Ultra-Secure Multi-Layer Hardware Access Control System (USMHACS)

- Designed a hardware-only access control system using combinational and sequential logic without microcontrollers.
- Implemented tamper detection, hardware lockout mechanisms, and dual-condition reset logic using decoders, counters, and SR latches.
- Simulated and verified system behavior using Proteus, analyzing intrusion scenarios and failure states.

ESP32-Based Multi-Functional Robotic System

- Developed an ESP32-based robotic platform integrating sensing, control and wireless communication.
- Implemented line-following, obstacle avoidance, and Bluetooth-based control modules.
- Designed a modular control architecture enabling reliable real-time task coordination.

Technical Skills

Programming: Python, C, C++

Machine Learning Interests: Computer Vision Systems, Biometric Authentication Systems, Deep Learning Fundamentals

Embedded Systems: ESP32, Arduino, Raspberry Pi, Sensor Integration (IR, Ultrasonic, Camera, RFID)

Digital Systems: Digital Logic Design, Decoders, Counters, SR Latches, Hardware Security Architectures

Simulation and Verification: Proteus 8/9, Cadence IMC

Communication Protocols: I2C, UART, SPI

———— Achievements and Programs

Aspire Leaders Program

Selected and completed an international leadership and professional development program led by Harvard faculty, focusing on leadership, problem-solving, and global collaboration.